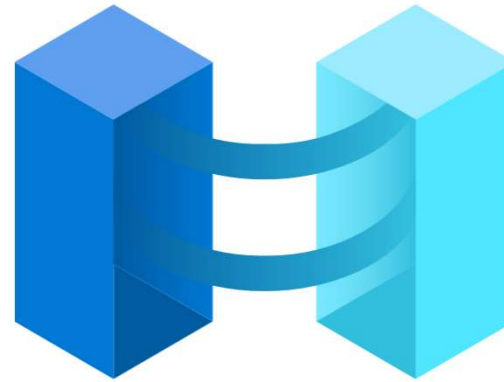


Secure, Govern, Automate: Der moderne Operation-Stack mit Azure Arc

Gregor Reimling





Gregor Reimling (he/him) th1

Chief Azure Technologist @ adesso SE, Germany

Gregor is awarded with the Microsoft MVP for Microsoft Azure and Security. He works as Chief Azure Technologist for adesso SE and is technical lead for Azure . His main areas are Microsoft Azure, Enterprise scale architectures, Cloud Security, Governance, Hybrid and Migration.



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[@GregorReimling](https://twitter.com/GregorReimling)



[/in/GregorReimling](https://in/GregorReimling)



youtube.com/@GregorReimling



github.com/GregorReimling



Azure & Security

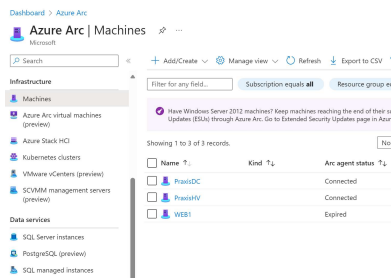
Slide 2

th1

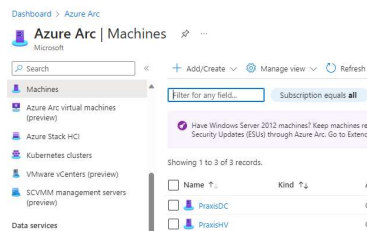
Sicher, dass du deine Pronomen nicht anpassen möchtest? =)

thorben.forke@adesso.de, 2025-02-20T00:03:28.494

Agenda



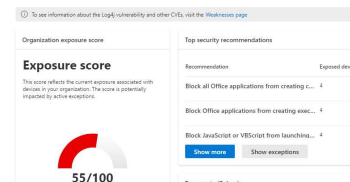
Overview about
Azure Arc



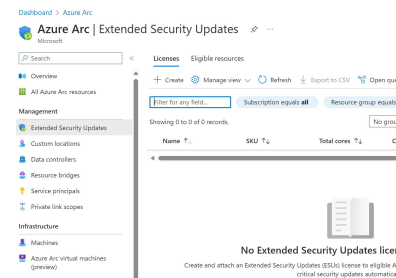
Server Management



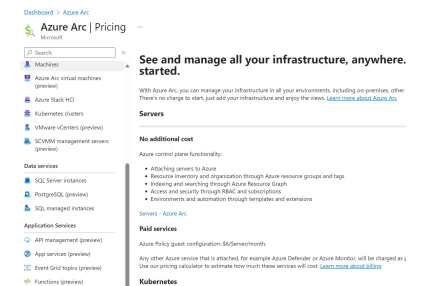
Microsoft Defender Vulnerability Management dashboard



Azure Automanage
Machine Configuration

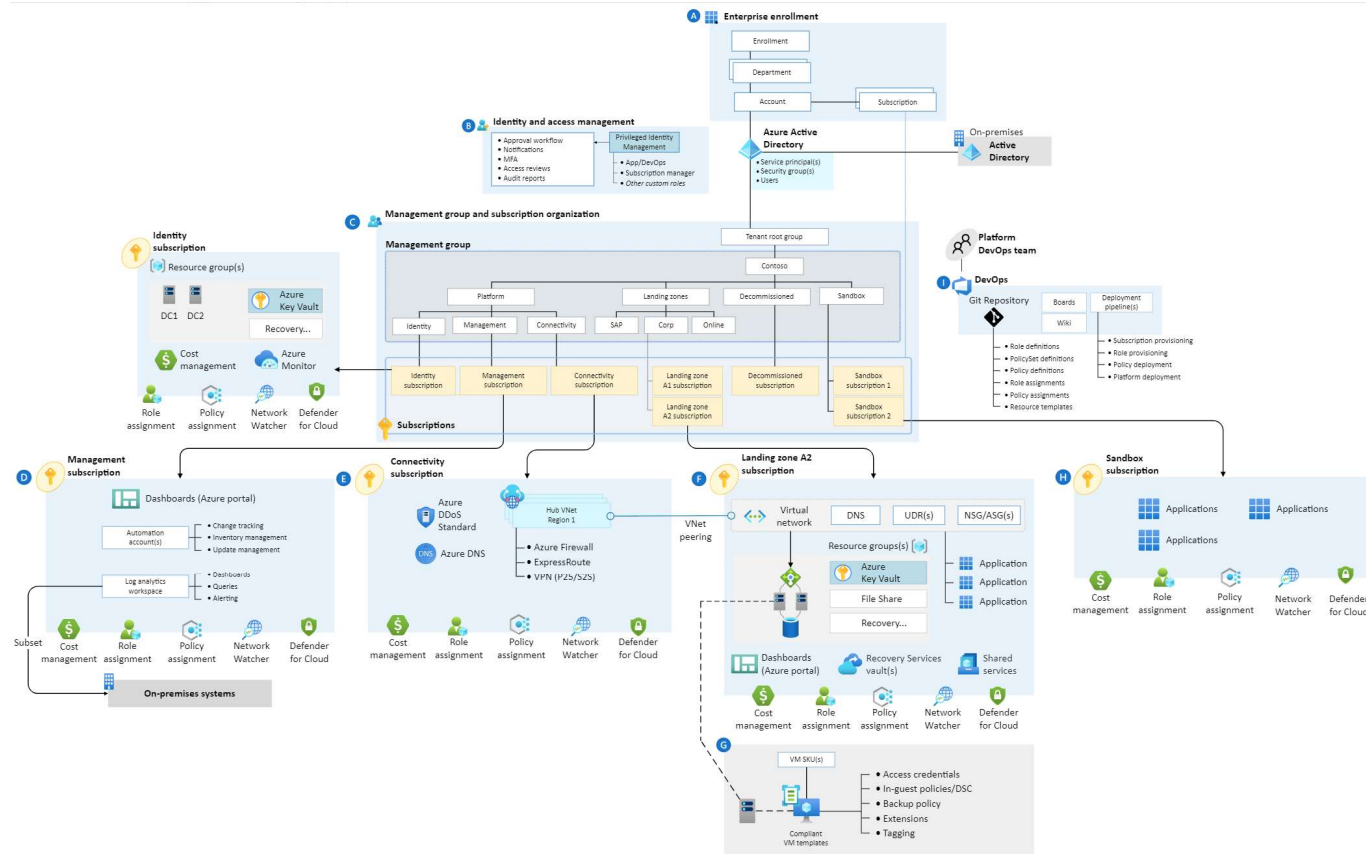


Update Management
Center

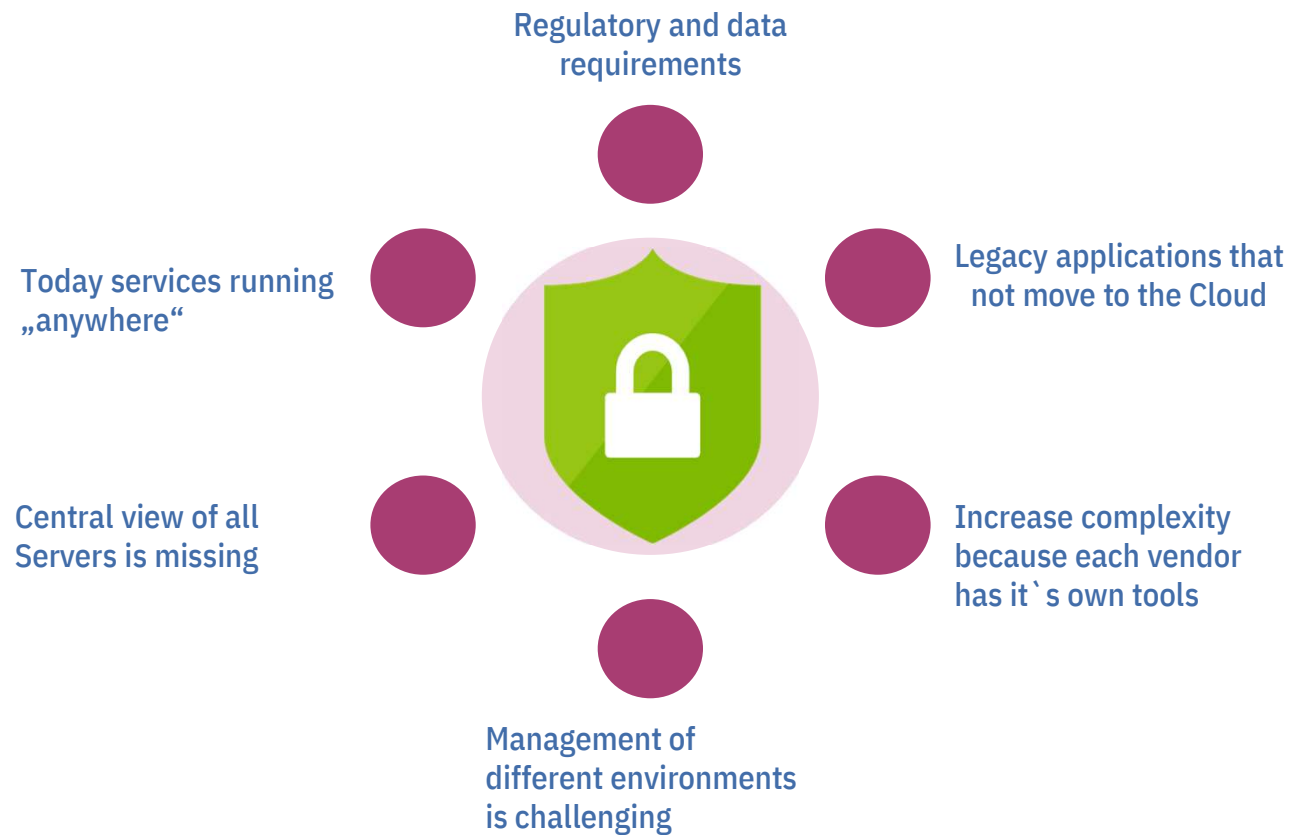


Defender for Server

Enterprise Scale



Reasons for Hybrid Infrastructures





Unified operations, management,
compliance, security and governance



Azure resources



Azure Arc-enabled infrastructure resources
(Servers, SQL servers, Kubernetes)



Azure Arc-enabled services resources
(Data services, App services, Machine Learning services)



Azure Resource
Manager



Azure Arc

Azure Arc-enabled
infrastructure onboarding

Azure Arc-enabled
services deployment

Azure Arc-enabled
infrastructure onboarding

On-premises IT
infrastructure resources



On-premises Arc-enabled services
(Data services, App services, Machine Learning services)



Azure Stack HCI

vmware

Multicloud Arc-enabled services
(Data services, App services, Machine Learning services)



Amazon Web Services



Google Cloud Platform

Multicloud IT
infrastructure resources



Azure Portal

CLI

Infra as Code

PowerShell



Azure Resource Manager

Inventory and organization



Azure Resource
Graph



Management Groups



Resource Groups



Service
Groups



Tags

Configuration management



Azure Update
Manager



Machine Configuration



Azure
Policy

Security & observability



Microsoft
Entra ID



Microsoft
Defender



Microsoft
Sentinel



Azure
Monitor

Infrastructure-as-Code & Deployments



Bicep
Templates



Azure Template Specs



Deployment
Stacks

Operations & management services

Azure Arc



Managed Servers, Kubernetes & Serverless Containers anywhere

Public Cloud

Hybrid Cloud

Multi-cloud

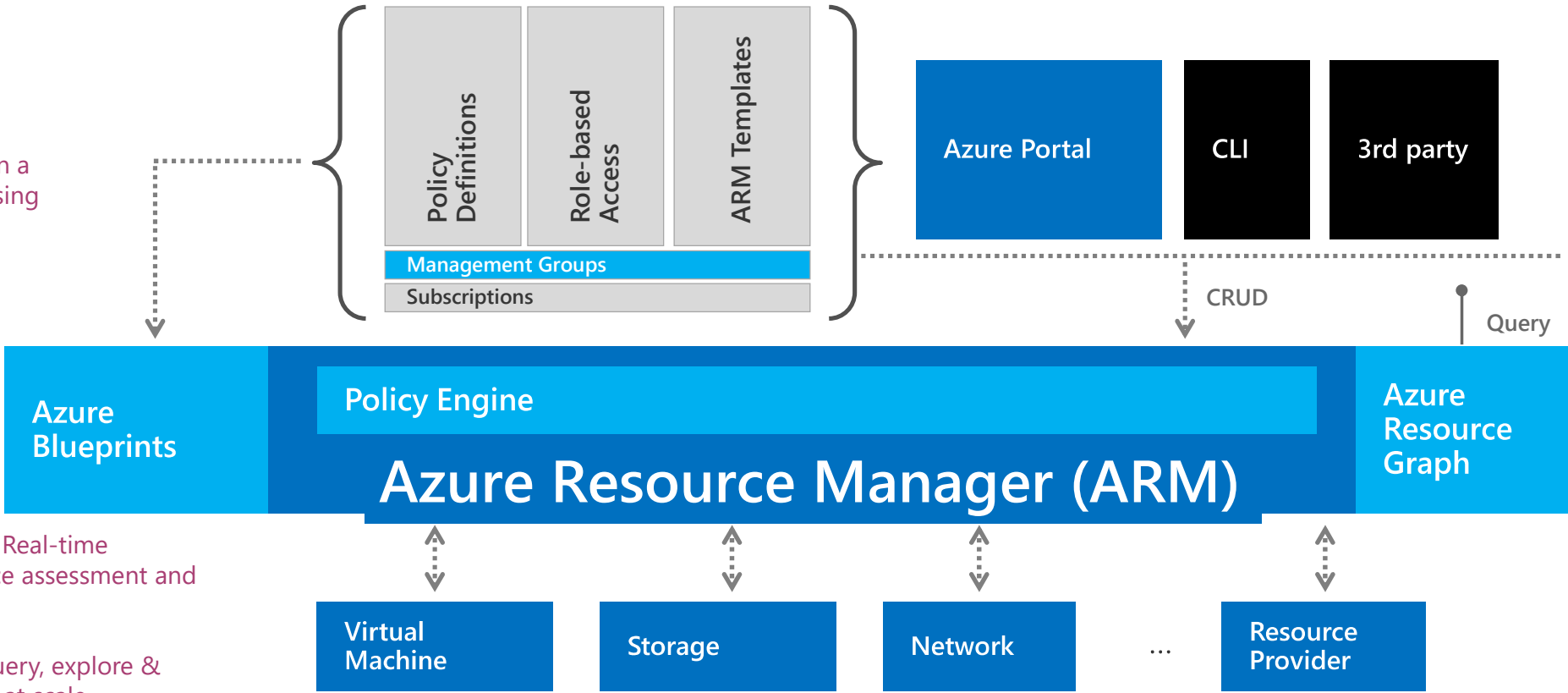
Edge

IoT

Azure Governance Architecture

providing control over the cloud environment, without sacrificing developer agility

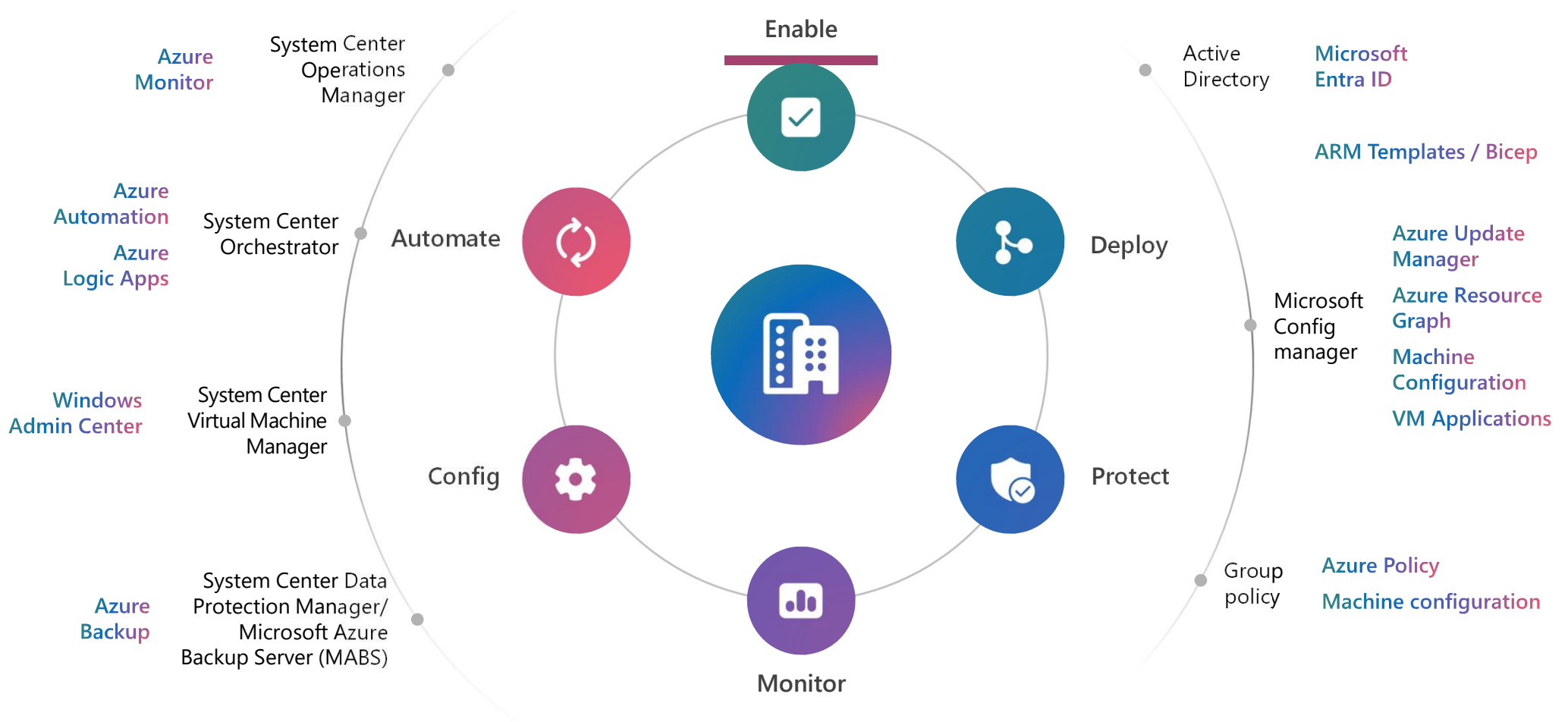
1. Environment Factory:
Deploy and update
cloud environments in a
repeatable manner using
composable artifacts



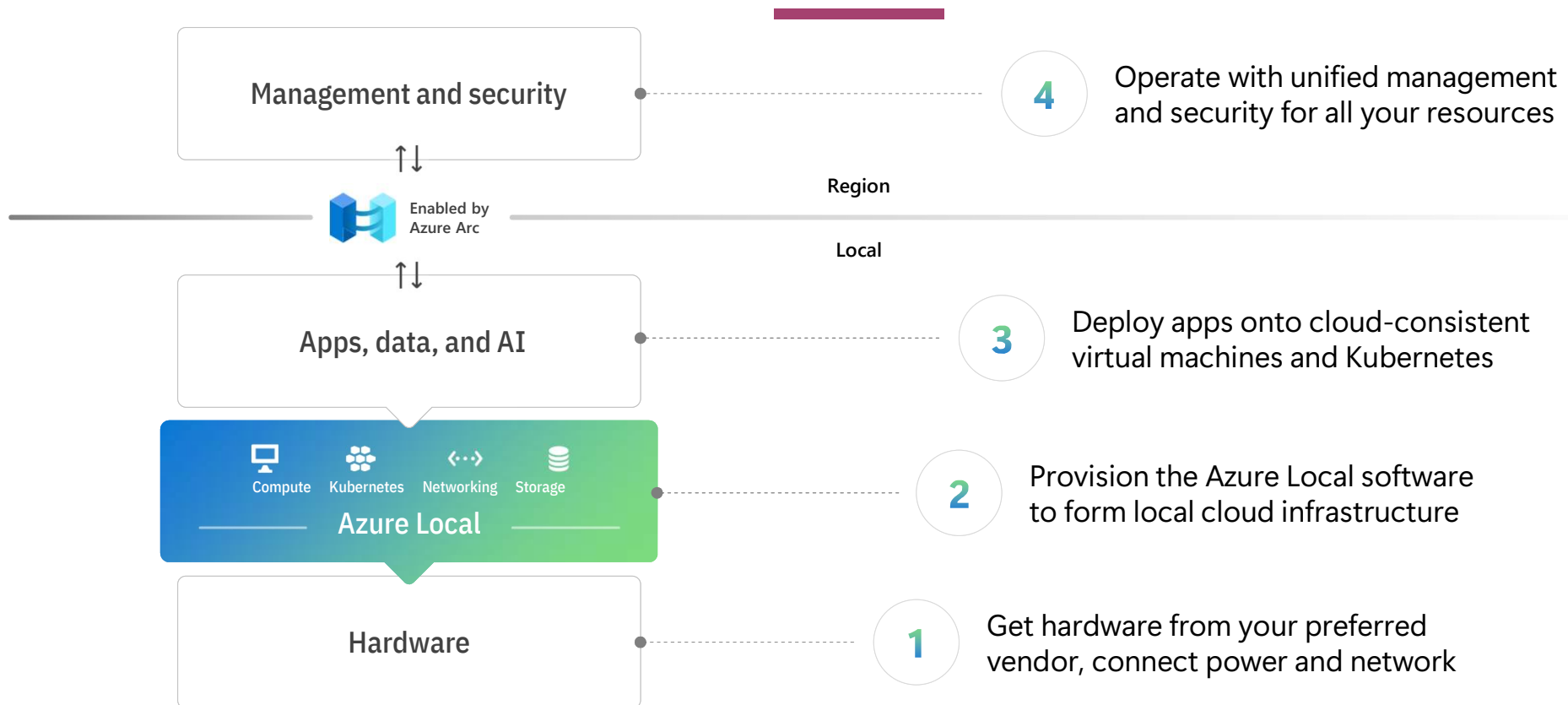
2. Policy-based Control: Real-time
enforcement, compliance assessment and
remediation at scale

3. Resource Visibility: Query, explore &
analyze cloud resources at scale

Azure's strategy for managing hybrid environments



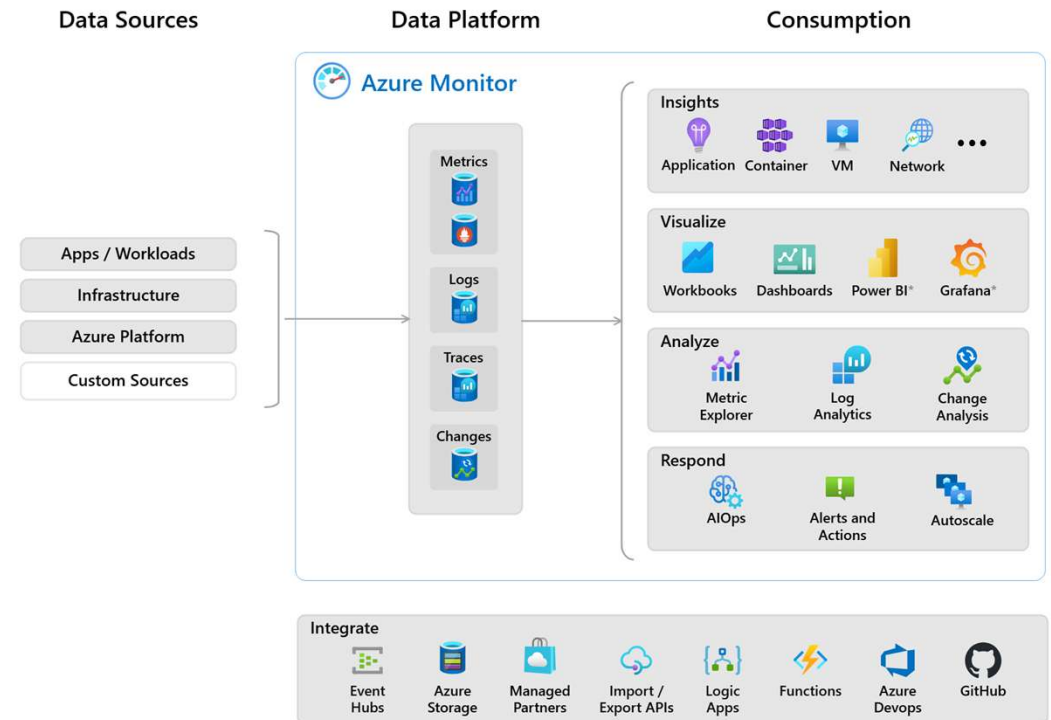
How Azure Local works (connected)



Monitor anywhere with Azure Monitor enabled by Azure Arc

Consistent monitoring across your hybrid and multi-cloud compute

- ➔ Detect & diagnose issues across **apps and dependencies** with application insights
- ➔ Correlate issues at **infra level** with insights for VMs, containers, storage, network, etc.
- ➔ Operationalize at scale with smart **alerts** and automated **actions**
- ➔ Drill down with **Log Analytics** for troubleshooting & deeper diagnostics
- ➔ Create **visualizations** with workbooks, Azure Monitor dashboards with Grafana, and Azure Managed Grafana



Supported Environments and OS



Environments

- VMware (including Azure VMware)
- Azure Stack HCI
- GCP, AWS, etc.



Windows

- Windows Server 2008 R2 SP1 and higher (including Core)
- Windows IoT Enterprise



Linux

- Ubuntu 16.04, 18.04, 20.04 and 22.04
- Debian 10 and 11
- CentOS Linux 7 and 8
- Rocky Linux 8
- SLES 12 and 15
- RHEL 7 and 8
- Amazon Linux 2
- Oracle Linux 7 and 8

Prerequisites

NET Framework 4.6

Windows PowerShell 4 (included in WS2012R2 and higher)

Azure RBAC

- **Onboarding:** Azure Connected Machine Onboarding
- **Read, Modify, Delete:** Azure Connected Machine Ressource Admin

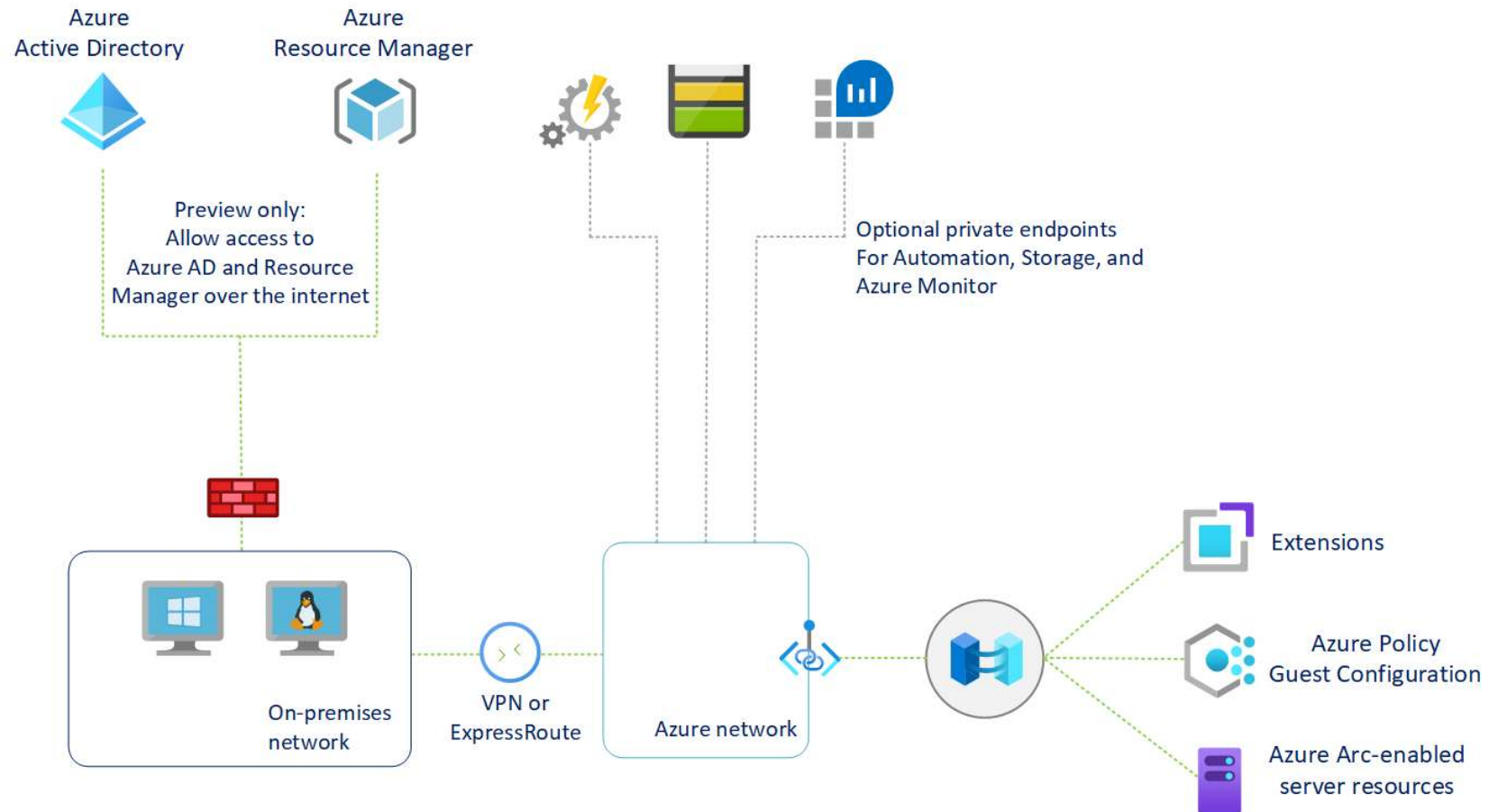
Resource Providers

- **Microsoft.HybridCompute**
- **Microsoft.GuestConfiguration**
- **Microsoft.HybridConnectivity**

Outbound via TCP 443 (Proxy server is supported)

Private Link support

Private Link



[Use Azure Private Link to securely connect servers to Azure Arc - Azure Arc | Microsoft Learn](#)

Connecting VMs

```
PS C:\Users\Administrator> try {
    $env:SUBSCRIPTION_ID = "009c17ca--4905999fba2d";
    $env:RESOURCE_GROUP = "arc_rg";
    $env:TENANT_ID = "e4f80c4f--3141bca1ced3";
    $env:LOCATION = "westeurope";
    $env:AUTH_TYPE = "token";
    $env:CORRELATION_ID = "95256887-01a2-4c82-948e-e457830cda97";
    $env:CLOUD = "AzureCloud";
    [Net.ServicePointManager]::SecurityProtocol = [Net.ServicePointManager]::SecurityProtocol -bor 3072;
    # Download the installation package
    Invoke-WebRequest -UseBasicParsing -Uri "https://aka.ms/azcmagent-windows" -TimeoutSec 30 -OutFile "$env:TEMP\install_windows_azcmagent.ps1";
    # Install the hybrid agent
    & "$env:TEMP\install_windows_azcmagent.ps1";
    if ($LASTEXITCODE -ne 0) { exit 1; }
    # Run connect command
    & "$env:ProgramW6432\AzureConnectedMachineAgent\azcmagent.exe" connect --resource-group "$env:RESOURCE_GROUP" --tenant-id "$env:TENANT_ID" --location "$env:LOCATION" --subscription-id
"$env:SUBSCRIPTION_ID" --cloud "$env:CLOUD" --tags "Datacenter=Ohligs, City=Solingen, StateOrDistrict=NRW, CountryOrRegion=Germany, Service=Arc, Environment=Prod" --correlation-id
"$env:CORRELATION_ID";}
catch {$logBody =
@{subscriptionId="$env:SUBSCRIPTION_ID"; resourceGroup="$env:RESOURCE_GROUP"; tenantId="$env:TENANT_ID"; location="$env:LOCATION"; correlationId="$env:CORRELATION_ID"; authType="$env:AUTH_TYPE
"; messageType=$_FullyQualifiedErrorId; message="$_";};
    Invoke-WebRequest -UseBasicParsing -Uri "https://gbl.his.arc.azure.com/log" -Method "PUT" -Body ($logBody | ConvertTo-Json) | out-null;
    Write-Host -ForegroundColor red $_.Exception;}
VERBOSE: Installing Azure Connected Machine Agent
VERBOSE: .NET Framework version: 4.6.1586
VERBOSE: Downloading agent package from https://aka.ms/AzureConnectedMachineAgent to C:\Users\ADMINI~1\AppData\Local\Temp\AzureConnectedMachineAgent.msi
VERBOSE: Installing agent package

Installation of azcmagent completed successfully
time="2022-11-11T22:16:46+01:00" level=info msg="The computer is connected in Azure. This may take a few minutes."
time="2022-11-11T22:17:59+01:00" level=info msg="Log in using the pop-up browser to authenticate yourself."
```



Why Azure Arc Gateway

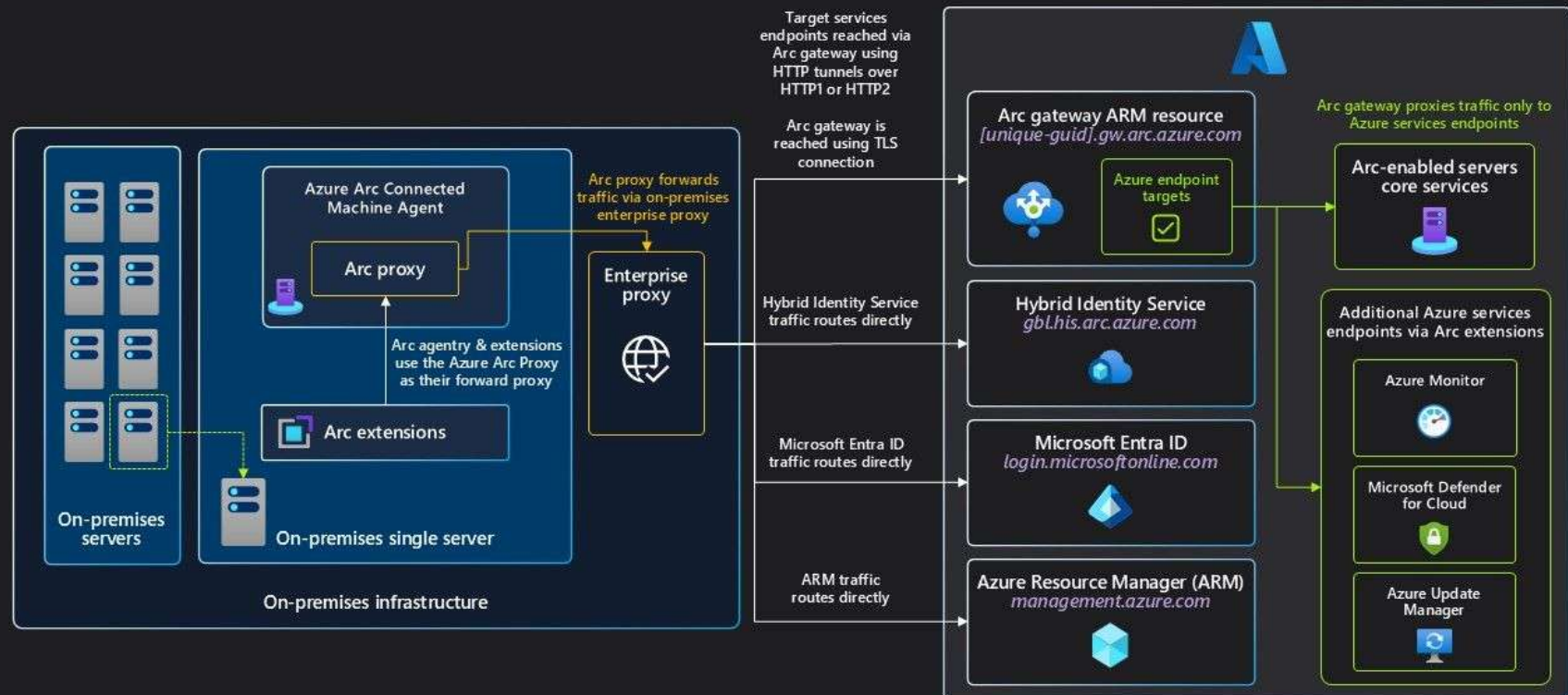
- Azure Arc Gateway provides a single endpoint for all inbound traffic from Arc-enabled resources
- **Endpoints required without Arc gateway = 17**
 - Using Extensions can grow up this numbers up to 150+ endpoints
 - Each extension has his own
- **Endpoints required without Arc gateway = 8**
- **Arc Gateway has two components**
 - Arc Gateway – Common Frontend for Arc traffic. Gateway is served on a specific domain
 - Arc Proxy – Component that routes all Arc traffic to it 's destination in Azure



Azure Arc Gateway

Azure Arc gateway – Arc-enabled servers solution architecture (Public Preview)

Powered by
Arc Jumpstart



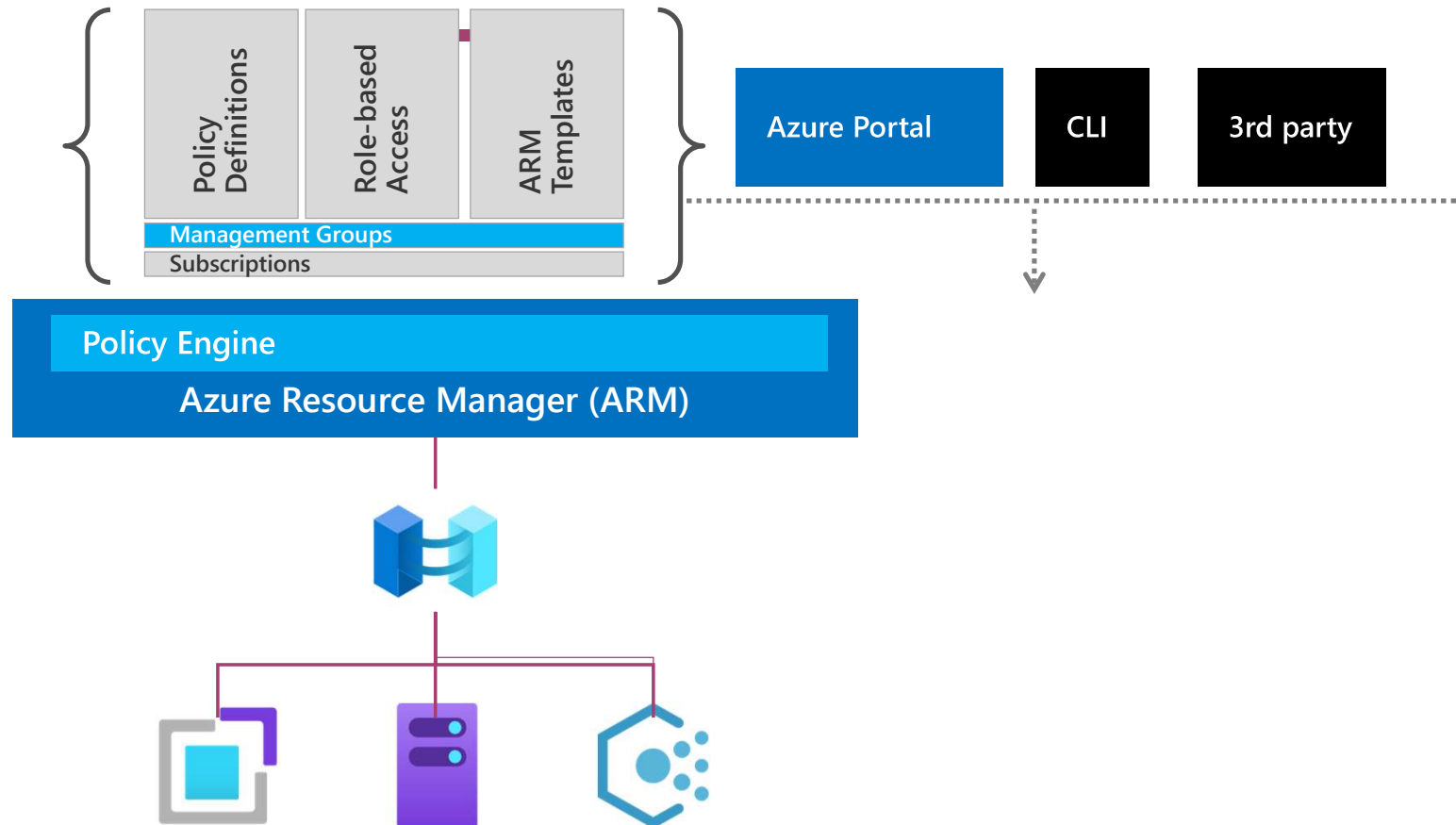
Azure Architecture

providing control over the cloud environment, without sacrificing developer agility

1. Environment Factory:
Deploy and update
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2. Policy-based Control: Real-time
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3. Resource Visibility: Query, explore &
analyze cloud resources at scale



Connected Machine Agent



Azcmagent tool configure the Azure Connected Machine agent



Updates coming via Windows Update (configure is mandatory)

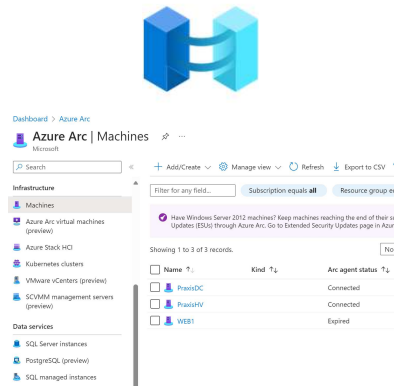


Server Renaming does not rename the Azure resource name (because is immutable)

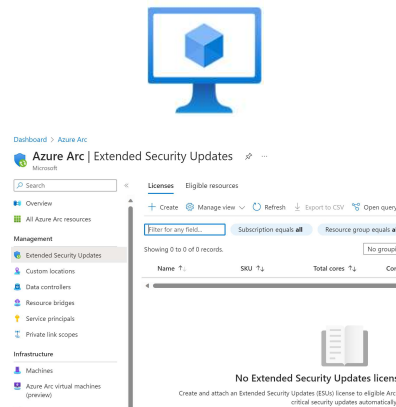
Delete and re-create is needed for Server renaming
Remove any VM-extension
Use azcmagent to disconnect

Windows Server doesn't check for updates in Microsoft Update by default.
! Configure Windows Update client to check for other Microsoft products !

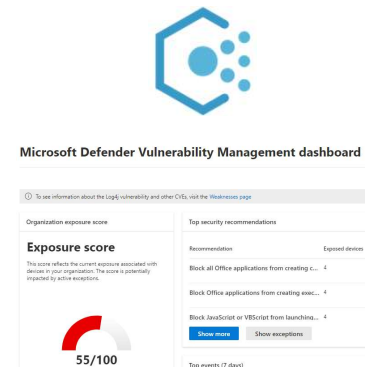
Demo Azure Arc



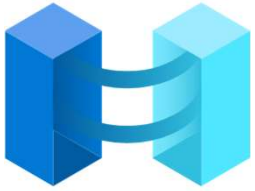
Overview about
Azure Arc



Server Management



Azure Automanage
Machine Configuration



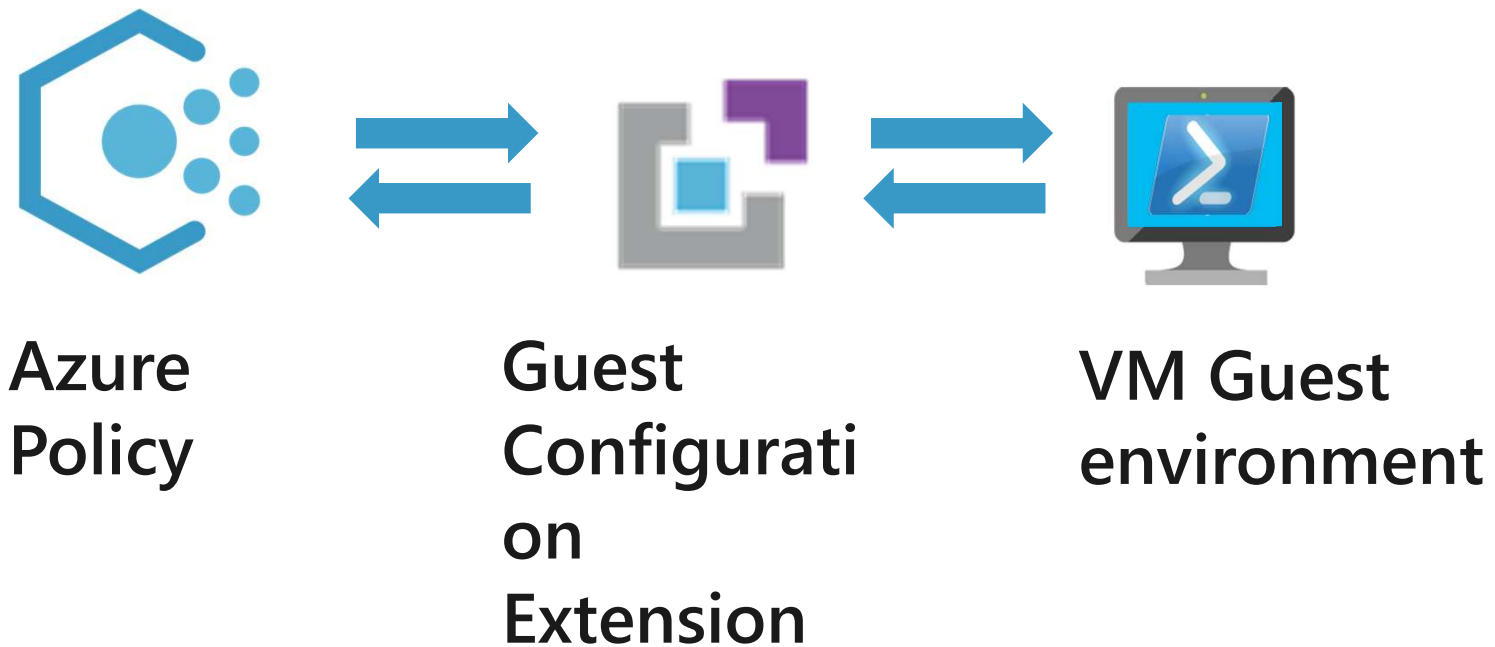
Azure Policy Guest Configuration

Renamed to

Azure Automanage Machine Configuration

Coming soon: guest configuration renames to machine configuration -
Microsoft Community Hub

How VM guest policy works



Guest Assignments

[Dashboard](#) >

Guest Assignments ...

Build Clouds

 Create  Manage view   Refresh  Export to CSV  Open query |  Assign tags  Delete

Subscription equals **all** Resource group equals **all**  Location equals **all**   Add filter

No grouping

☐	Name 	Machine 	Type 	Status 	Resource
☐	 AuditSecureProtocol (W...	WEB1	Microsoft.HybridCompute	NonCompliant	arc_rg
☐	 AzureWindowsBaseline ...	WEB1	Microsoft.HybridCompute	NonCompliant	arc_rg
☐	 WindowsDefenderExplo...	WEB1	Microsoft.HybridCompute	Compliant	arc_rg
☐	 WindowsLogAnalyticsA...	WEB1	Microsoft.HybridCompute	Compliant	arc_rg



Azure Automanage Machine Configuration

Guest configuration extend
Azure Policy to Server

Perform audit and
configuration inside Server

Need Resource Provider
Microsoft.GuestConfiguration

Checks for changes every 5
minutes

Installs security baselines for
Windows and Linux

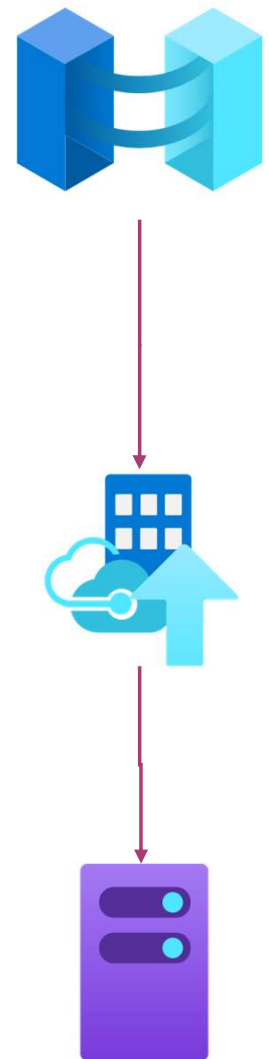
Configured in audit-only
mode

Non-compliant devices are
displayed but not reset

Reset possible through
advanced configuration



Azure Update Manager



Patch Orchestration

Azure Managed – Safe Deployment

- Supported for Linux und Windows
- This mode activate automatic patching of the VM
- Assessment will be stored in Azure Resource Graph
- Support Patches by Availability

Customer Managed Schedules

- Only for Windows VMs
- Support no Patches by Availability
- Default Mode when nothing is configured

Windows automatic updates

- Configuration of the Windows VM would be used (Registry-/GPO-settings)

Manual updates

- Only for Windows VMs
- Support no Patches by Availability
- Mode only be used when 3rd party patch solution is in place

ImageDefault

- Only for virutal Linux VMs
- Default Mode when nothing is configured
- Support no Patches by Availability

Hotpatch



WS 2022 Datacenter: Azure Edition Server Core

Azure = GA

Azure Stack HCI = GA

Hotpatch activated by default



WS 2022 Datacenter: Azure Edition mit Desktop

Azure = GA

Azure Stack HCI = GA



Hotpatching for Azure Arc

Support for Windows Server 2025

(Standard and Enterprise is available)

Status for older OS Systems not known

[Hotpatch für Windows Server: Azure Edition | Microsoft Learn](#)

[Hotpatching: Improving server security and productivity | Windows Server Summit 2024](#)

Price Overview

Service	Price
Azure	Free
Extended Security Updates (ESUs) via Arc	Free
Defender for Server P2 via Arc	Free
Azure Stack HCI clusters	Free
Azure Arc	5\$ (4,62€)
Other enabled Defender Plans via Arc	0,16\$ per Day (5\$ per Month)



Price changes with Windows Software Assurance

- Microsoft announced on Ignite 2024 that the following services are free to customers with activated Windows Server Software Assurance
 - Azure Update Manager
 - Azure Change Tracking and Inventory
 - Azure Machine Configuration
 - Windows Admin Center in Azure for Arc
 - Remote Support
 - Network HUD
 - Best Practices Assessment
 - Azure Site Recovery (Configuration Only)

Azure benefits

 It is recommended that you upgrade the Azure Arc agent on this machine to at least version 1.48 to claim Azure benefits.

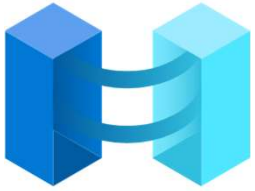
If your machine license is covered by active Software Assurance, you can access key management services at no additional cost. [Learn more](#) 

Activate Azure benefits

☐ By checking this box, you attest that your Windows Server licenses have active Software Assurance or your Windows Server licenses are active subscription licenses.

 Cannot claim Azure benefits because machine must be licensed to be eligible.

Announcing General Availability: Windows Server Management enabled by Azure Arc | Microsoft Community Hub



Defender for Cloud

Microsoft Defender for Cloud



Continuously Assess

Know your security posture.
Identify and track vulnerabilities.



Secure

Harden resources and services with
Azure Security Benchmark and
AWS Security Best Practices standard



Defend

Detect and resolve threats to
resources and services.



[Defender for Azure Cosmos DB](#)



[Defender for Containers](#)

[Defender for App Service](#)

[Defender for DNS](#)

[Defender for CSPM](#)

[Defender for Servers](#)

[Defender for Key Vault](#)

[Defender for Resource Manager](#)

[Defender for Storage](#)



[Defender for SQL](#)

Activation of Defender for Servers

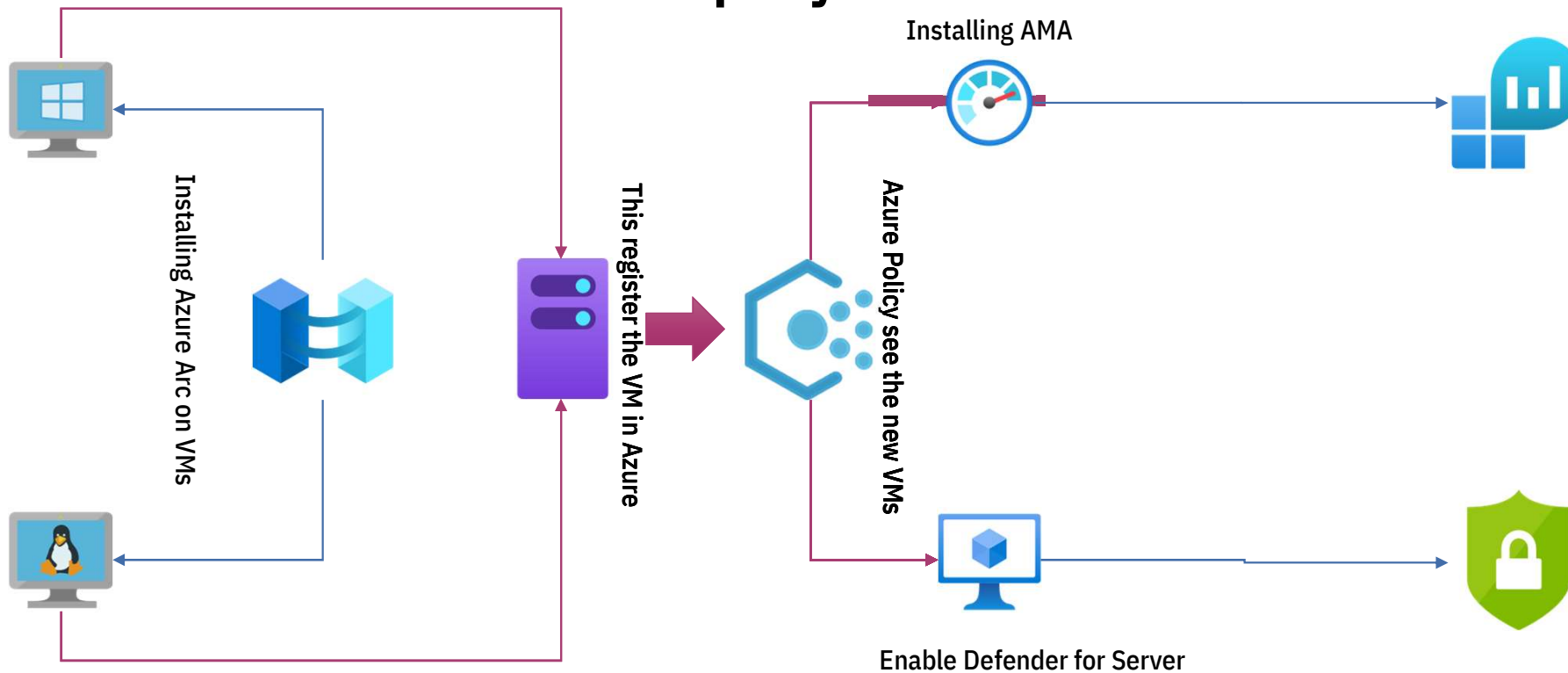
- Defender for Servers plan 1 must be enabled on subscription level
- Defender for Servers plan 2 must be enabled on subscription and **Workspace** level
- Mixing of the plans only possible with different subscription
- License cost for plan 2 is incurred for each machine connected to the Workspace where plan 2 is activated



Defender for Server

	Plan 1	Plan 2
Unified View	✓	✓
Automatic MDE provisioning	✓	✓
MS Threat and Vulnerability management	✓	✓
Security Policy and Regulatory Compliance		✓
Integrated Vulnerability by Qualys		✓
Log Analytics 500MB free data ingestion per day		✓
Threat detection		✓
Adaptive application control		✓
File integrity monitoring		✓
Just-in-Time VM access		✓
Adaptive Network hardening		✓
Docker host hardening		✓
Fileless attack detection		✓
Price	5\$ per Server	15\$ per Server

Deployment at Scale



- Deploy Microsoft Defender for Endpoint agent on Windows virtual machines
- Deploy Microsoft Defender for Endpoint agent on Windows Azure Arc machines
- Deploy Microsoft Defender for Endpoint agent on Linux hybrid machines
- Deploy Microsoft Defender for Endpoint agent on Linux virtual machines

What's new

Arc-enabled servers

Arc server agent auto-upgrade

Machine configuration –
OS policy editor

Customizable baselines

Windows Server 2025
Recovery Environment
configuration through policy

Arc-enabled Kubernetes

Access any other resources
in Azure securely with
workload identity

Offline access to secrets
on edge workloads

Managed ArgoCD Support

Multi-cloud management

Google Cloud Platform support

AWS S3 to Azure Storage mover

<http://aka.ms/ignite25/ArcBlog>

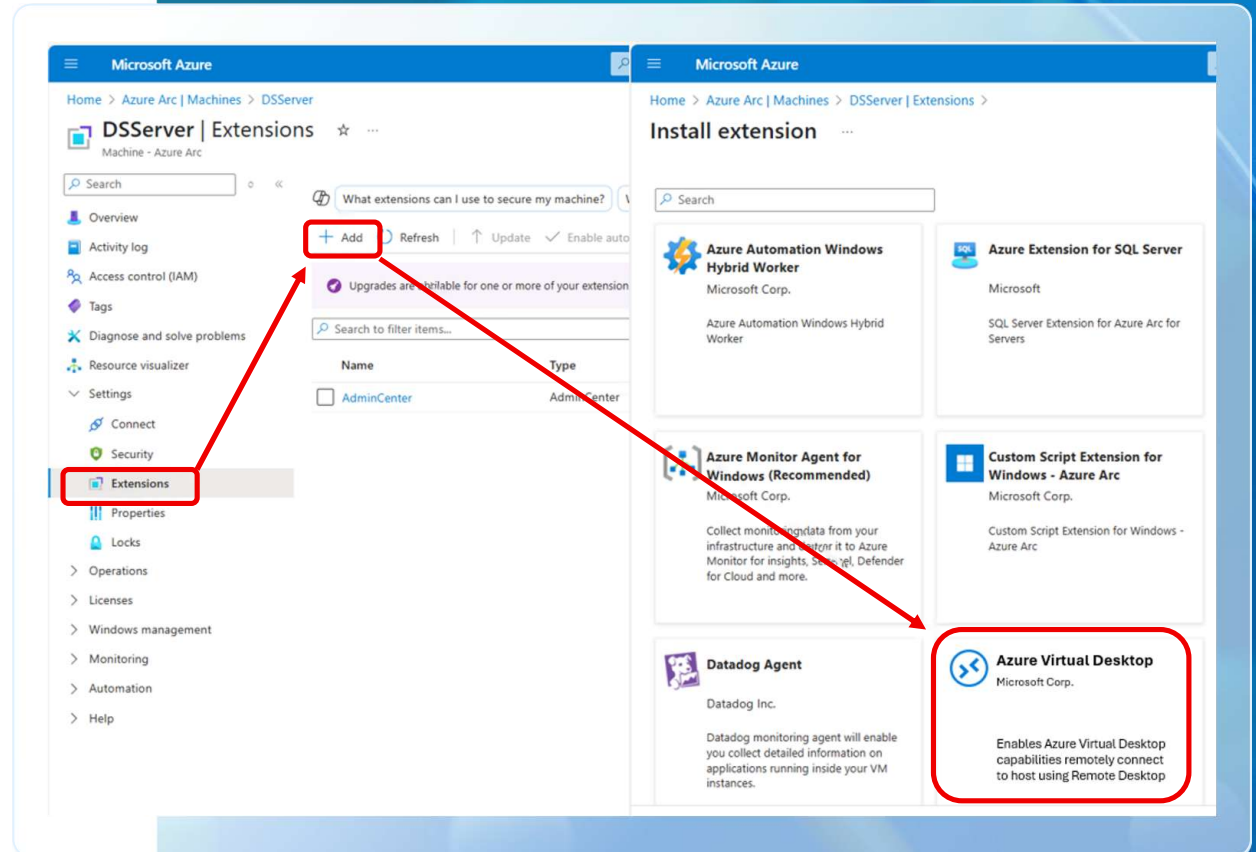
Azure Arc Azure Virtual Desktop extension deployment

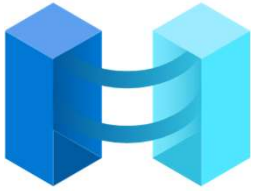
Pre-requisites

- Admin creates host pool
- Admin creates VM using tools of their choice

1. Install Azure Arc Connected Machine Agent
2. “Add” Azure Virtual Desktop Arc extension
 - Installs Azure Virtual Desktop agents
 - Installs Remote Desktop Session Host (RDSH role (Windows Server only)
 - Adds device to Azure Virtual Desktop host pool

Leverage Partner solutions for creating virtual machines (VM), lifecycle and power management





How To Start





Overview | Azure Arc Jumpstart

Jumpstart
Scenarios



Want to explore multiple environments and see the full breadth of Jumpstart? Get automated zero-to-hero scenarios for Arc-enabled servers, Arc-enabled Kubernetes, and more.

[Browse scenarios >](#)

Jumpstart
Agora



Explore cloud-to-edge scenarios designed for specific industry needs. Get a full-stack deployment with dedicated guides to walk you through the process—plus the end-to-end user experience.

[Browse industry solutions >](#)

Jumpstart
ArcBox



Get a complete Azure Arc environment in just one click. Explore all the major capabilities of Azure Arc in a virtual, hybrid sandbox—all you need is an Azure subscription to get started.

[View ArcBox capabilities >](#)

Jumpstart
LocalBox



Want to try Azure Local? This is the tool for you. Get a dedicated Azure Local sandbox in one click—all you need is an Azure subscription to get started.

[View LocalBox capabilities >](#)

Jumpstart
Drops



No matter how big or small the contribution, create a Drops to help others explore, discover, and leverage development artifacts. You can also find quick deployment guides, useful code snippets, and more.

[Browse community artifacts >](#)

Jumpstart
Gems



Explore detailed technical diagrams of Azure technologies and end-to-end cloud scenarios, simplifying the complex architectures and workflows.

[View diagrams and assets >](#)

How to start

Microsoft Azure Arc Community Monthly Meetup

Overview

Once a month, the various Azure Hybrid Cloud product groups at Microsoft will hold a call to showcase new features, talk through important topics and engage in a Q&A regarding Azure Arc. The foundational goals of the call are highlighted below:

- Provide the Azure Arc community with product updates
- Host a short talk and/or demo on Azure Hybrid Cloud technologies and products technologies
- Collect feedback from the community on issues, blockers, use cases, and questions related to Azure Hybrid Cloud technologies and products

Azure Arc Community Monthly Meetup

[GitHub - microsoft/azure_arc_community: Public repository for hosting the Azure Arc Community content](#)

Contributors 3



likamrat Lior Kamrat



microsoftopensource Microsoft Open ...



csand-msft Chris Sanders



aka.ms/ArcCustomerSignup

Security Recommendations

Keep the Agent
updated

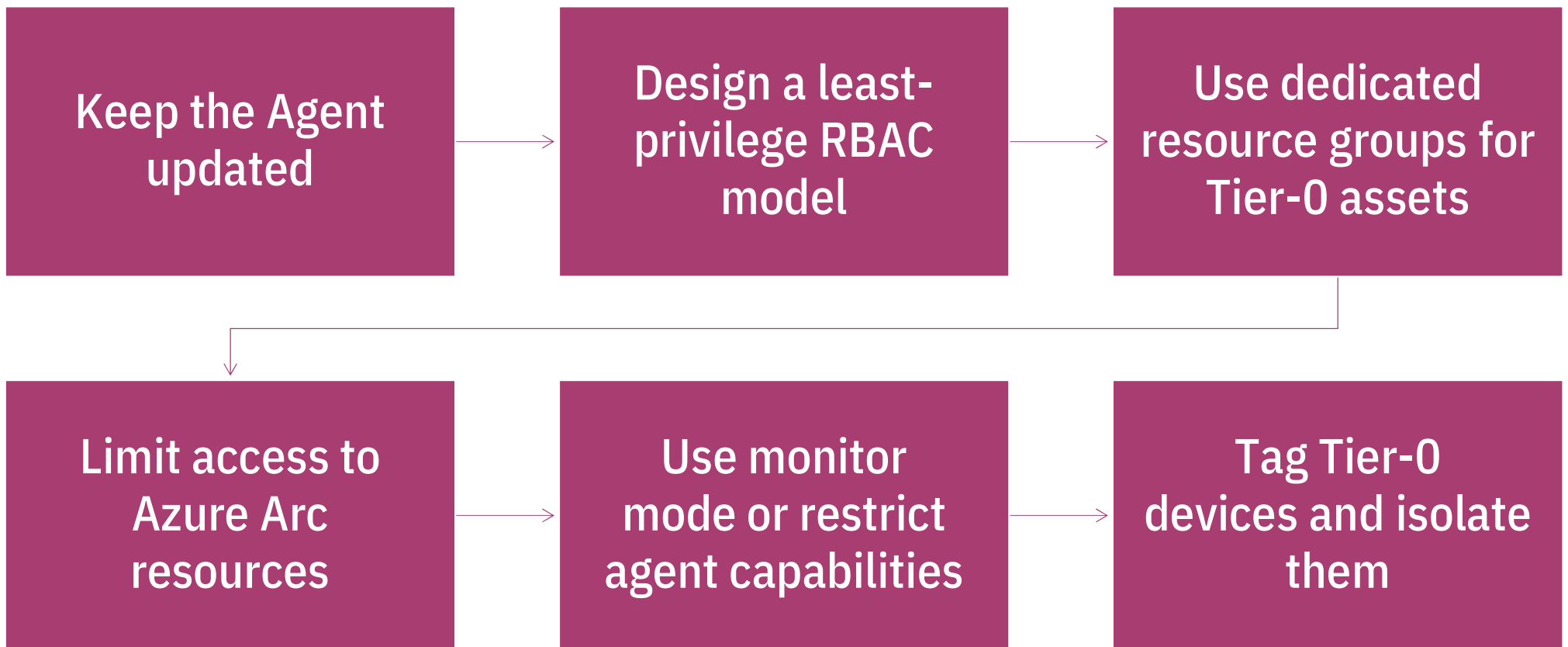
Design a least-
privilege RBAC
model

Use dedicated
resource groups for
Tier-0 assets

Limit access to
Azure Arc
resources

Use monitor
mode or restrict
agent capabilities

Tag Tier-0
devices and isolate
them



Summary and Best Practices

Before Start with Azure Arc think on Enterprise Scale and use a separate Hybrid Subscription

Group Arc resource in appropriate Resource groups and use Tags

Track Installation and Server Status with Workbooks

Always use AMA instead of MMA because of retirement of MMA

Think on additional pricing for Azure Policy in Guest configuration and Update Center

Integrate VMs outside of Azure with Arc to enable Defender for Servers everywhere



Links

- [Arc Jumpstart Training – YouTube](#)
- [A Guide to Adaptive Cloud at Microsoft Ignite 2024 | Microsoft Community Hub](#)
- [Azure Arc Jumpstart](#)
- [PowerShell Gallery | jumpstart](#)
- [Azure Arc Jumpstart Drops](#)
- [Build a Business case with Azure Migrate - Azure Migrate | Microsoft Learn](#)
- [We have been waiting for SO LONG...Arc Gateway!](#)
- [Azure Arc Security recommendation by Jeffrey Apel | LinkedIn](#)
- [Connect to AWS with the multicloud connector in the Azure portal - Azure Arc | Microsoft Learn](#)
- [Managing the Azure Arc-enabled servers agent - Azure Arc | Microsoft Learn](#)
- [Overview of the Azure Connected Machine agent - Azure Arc | Microsoft Learn](#)
- [Archive for What's new with Azure Arc-enabled servers agent - Azure Arc | Microsoft Learn](#)
- [GitHub - microsoft/azure_arc_community: Public repository for hosting the Azure Arc Community content](#)
- [Azure Automanage | Microsoft Learn](#)
- [Microsoft Cybersecurity Reference Architectures - Security documentation | Microsoft Learn](#)
- [Pricing – Azure Arc | Microsoft Azure](#)

Save the Date



Jetzt hafte ich

IT-Sicherheitsverantwortung 2026 – NIS2, Regulatorik
und persönliche Haftung

Speakers

Alexander Busse
Interim CISO & KI-Engineer
für Security Governance

Thomas Ullrich
Internationaler Management- &
Cybersecurity-Advisor für
Governance & Regulierung

Event

19. Februar 2026, 17:30 Uhr

Online & bei uns im Office
SThree Düsseldorf
Georg-Glock-Str. 12



Azure Arc – Person of Interests



Lior Kamrat

Principal Product Manager -
Azure Arc Platform

[Lior Kamrat | LinkedIn](#)



Thomas Maurer

~~Senior PM and Chief Evangelist~~
~~Azure Hybrid at Microsoft~~

[Thomas Maurer | LinkedIn](#)



Gregor Reimling (he/him) th1

Chief Azure Technologist @ adesso SE, Germany

Gregor is awarded with the Microsoft MVP for Microsoft Azure and Security. He works as Chief Azure Technologist for adesso SE and is technical lead for Azure . His main areas are Microsoft Azure, Enterprise scale architectures, Cloud Security, Governance, Hybrid and Migration.



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[@GregorReimling](https://twitter.com/GregorReimling)



[/in/GregorReimling](https://in/GregorReimling)



youtube.com/@GregorReimling



github.com/GregorReimling



Azure & Security

th1

Sicher, dass du deine Pronomen nicht anpassen möchtest? =)

thorben.forke@adesso.de, 2025-02-20T00:03:28.494